

Cancer Surveillance: From Data to Public Health Action

Core Concepts and Practice

Rauf Shah, MPH-EPI (Fall 2025)

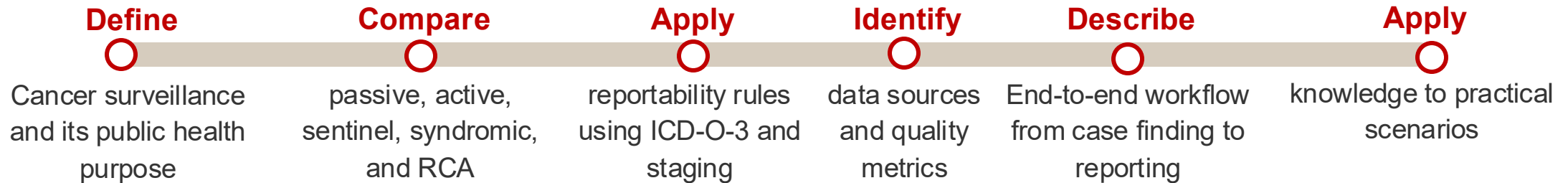
Course:

PUBLIC HEALTH TEACHING AND TRAINING



Learning Objectives & Agenda

Learning Objective



Agenda

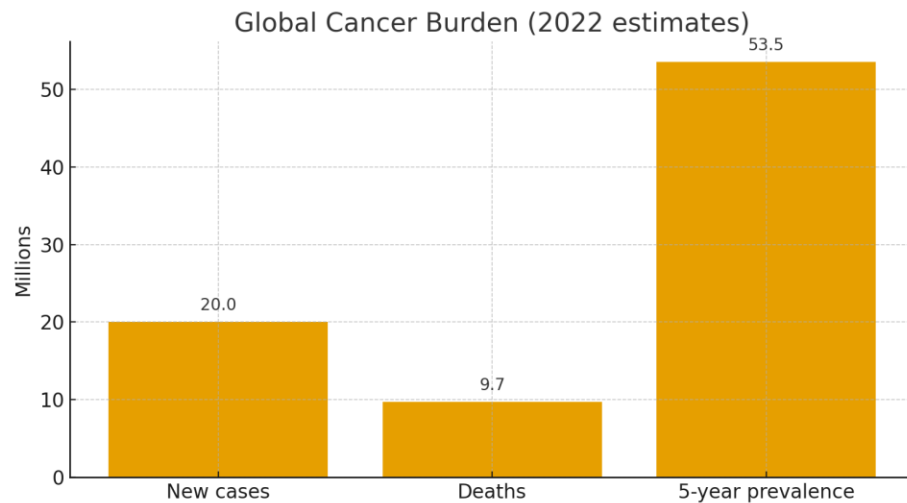
- ☐ 1. Introduction to Cancer Surveillance
- ☐ 2. Types & Definitions
- ☐ 3. Workflow & Reporting
- ☐ 5. Practical Applications & QA

Speaker Notes: These objectives map our journey from fundamentals to applied practice.

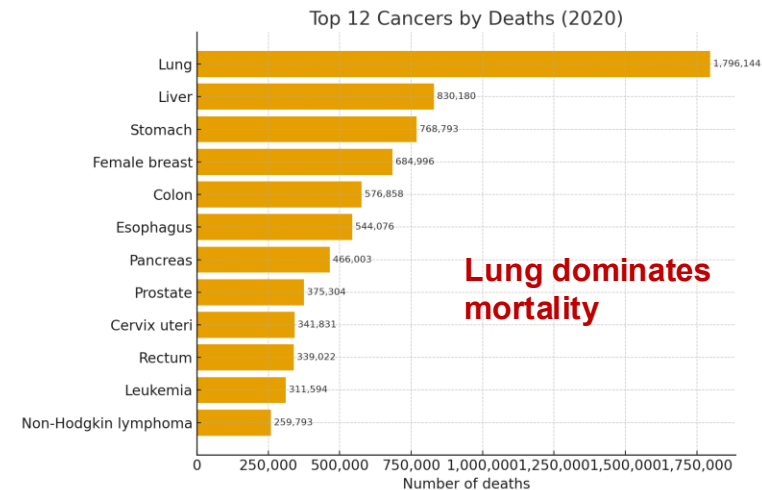
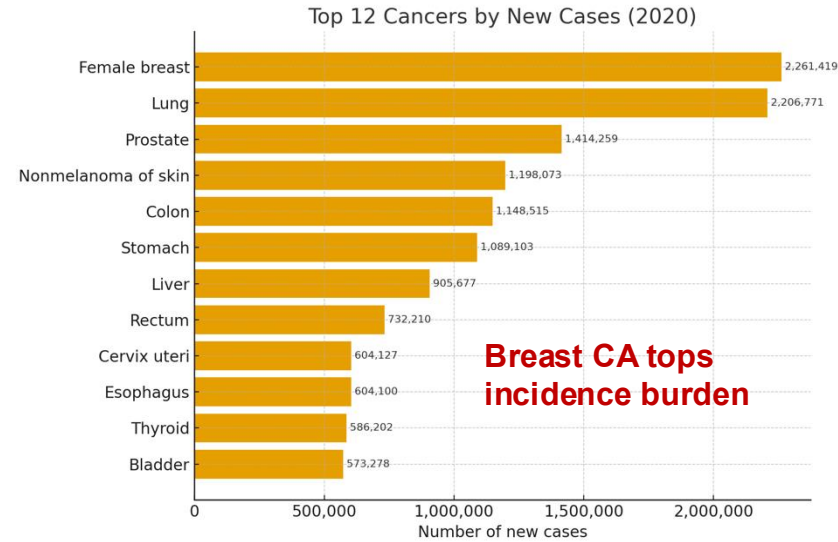
The Global Cancer Burden & Why Surveillance Matters

Key Global Insights

- 2022 estimates: ~20 million new cases and 9.7 million deaths worldwide; 53.5 million 5-year prevalence.
- Lung cancer ~12.4% of new cases globally; breast and colorectal follow.

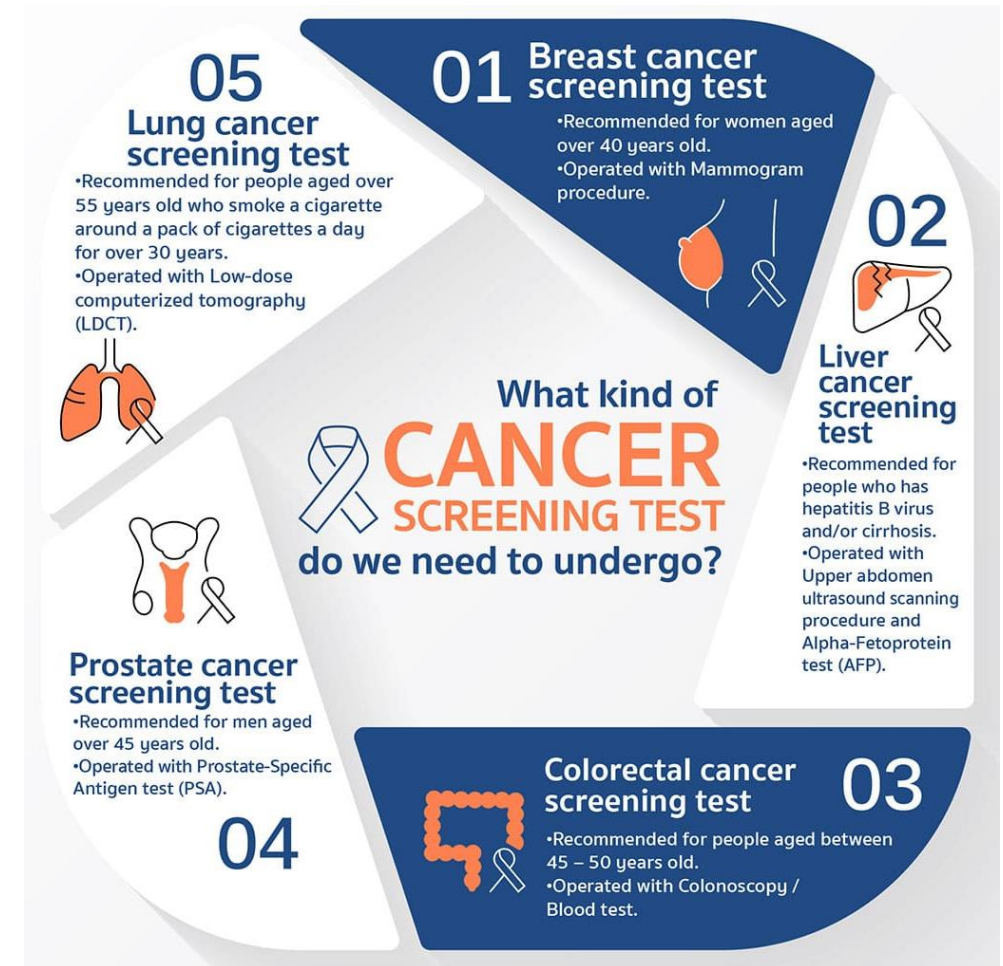


- These scale demand smart action.**
- Thus if we have a proper Surveillance System, it may turn counts into impact, targeted prevention, faster detection, thus saving millions of LIVES!**



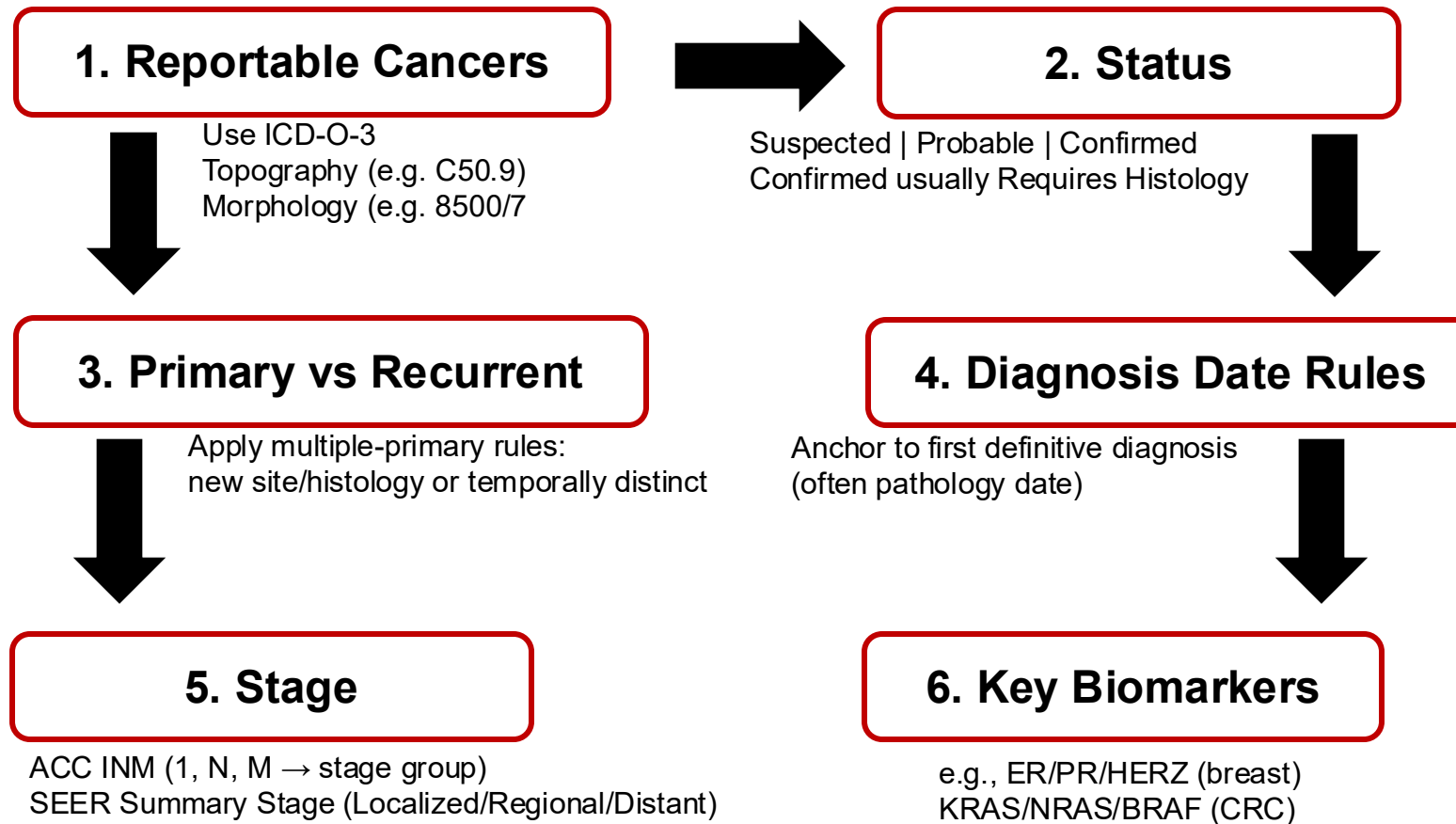
From Surveillance System Approaches To Screening Tests

Type	Mechanism	Best For	Challenges
Passive	Facilities submit reports	Broad coverage at scale	Variable data quality
Active	Registry staff proactively seek	Higher completeness	Resource-intensive
Sentinel	Selected high-quality sites	Rare cancers, rapid insight	Limited representativeness
Syndromic	Early signals (ED, labs, pharmacy)	Early warnings	Low specificity
RCA	Pathology-driven rapid reporting	Time-critical signals	Technical integration needs



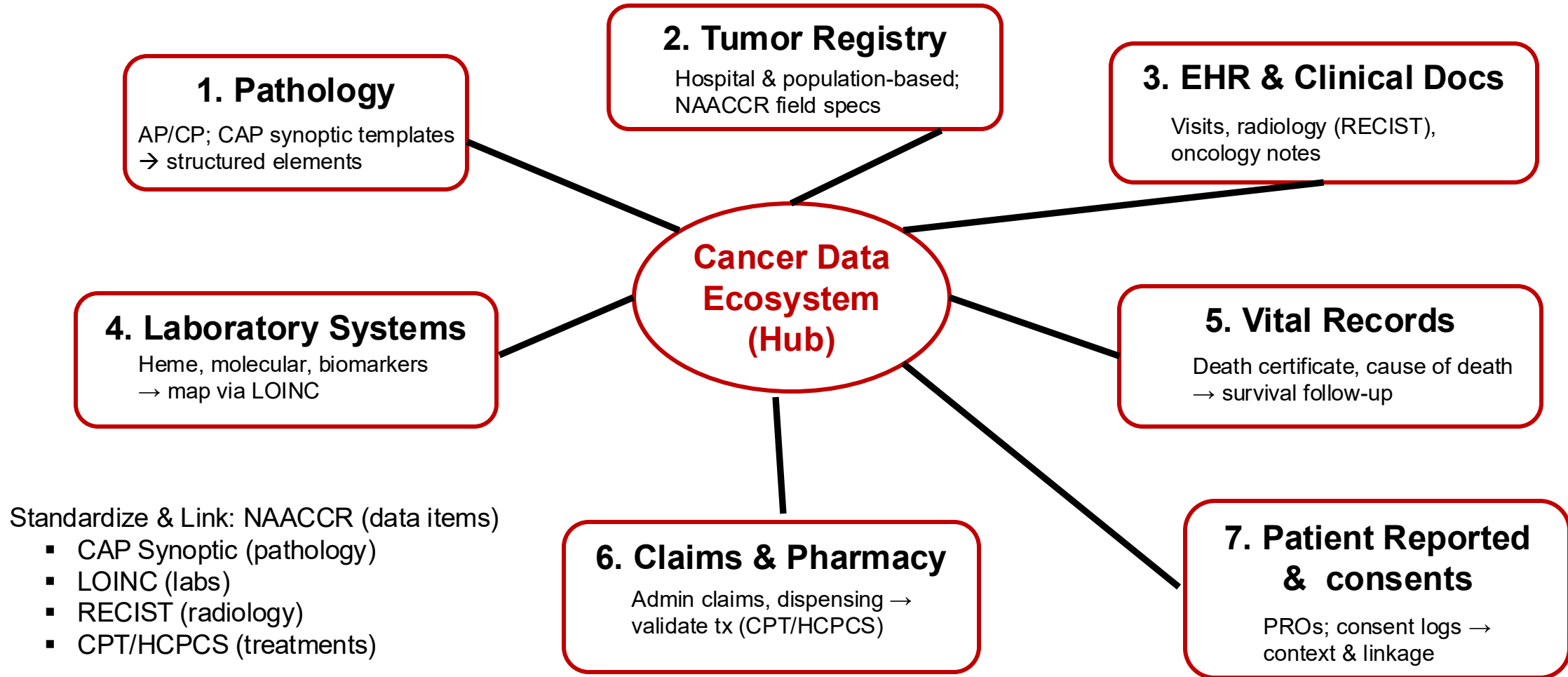
Right Data → Right Test → Right Time: Surveillance Turns Signals into Saved Lives.

Case Definitions & Classification



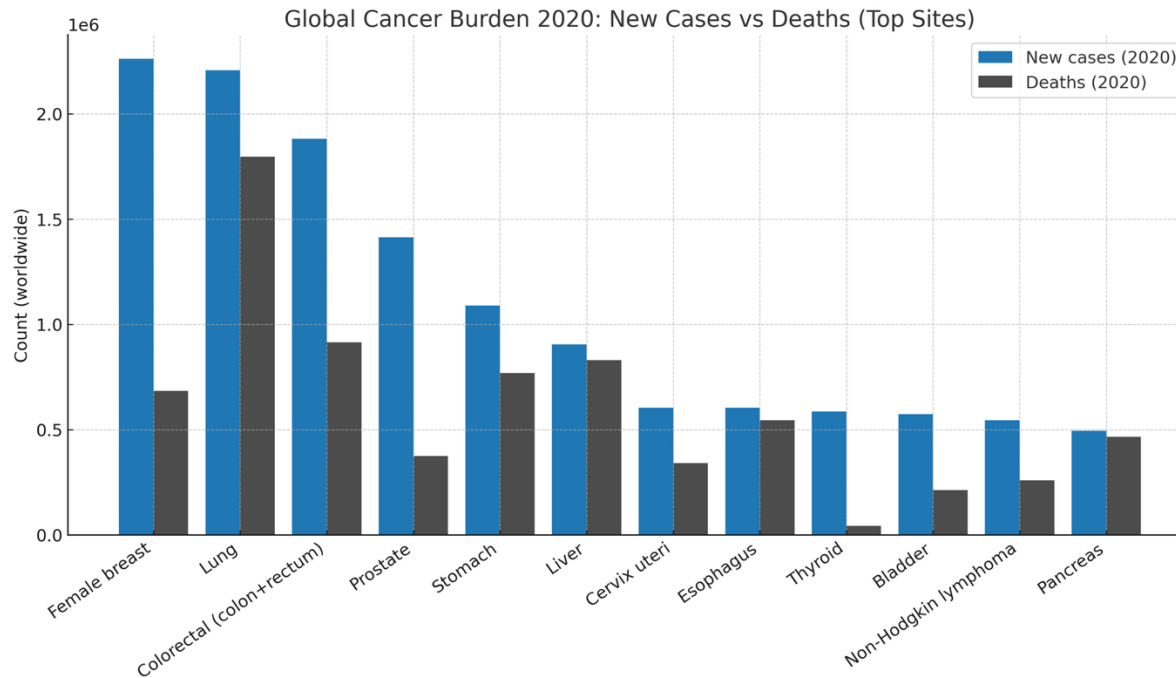
Get the case right, reportability, date, primaries, stage, and biomarkers, because every rate and policy built on it depends on that foundation.

Essential Data Sources



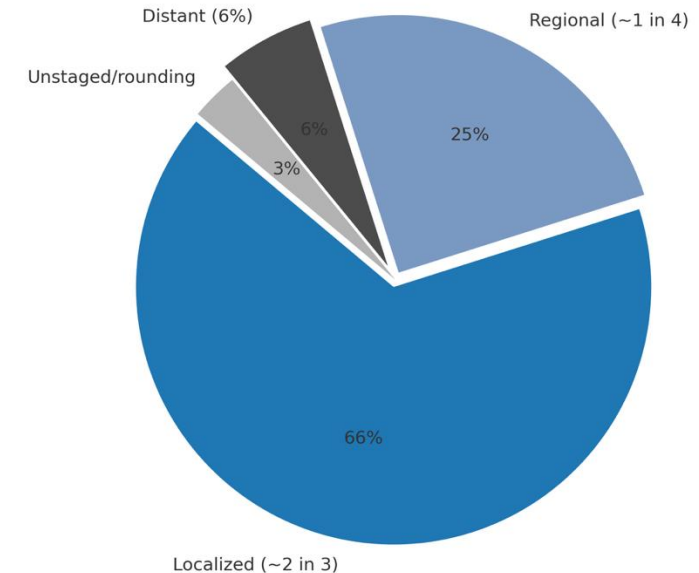
Turn many feeds into one truth: standardize (NAACCR/CAP/LOINC/RECIST/CPT) so scattered inputs become trusted population insight.

Core Parameters & Evaluation Indicators



Global Top Sites (2020): New cases vs. deaths - GLOBOCAN 2020

U.S. Female Breast Cancer — Stage at Diagnosis (2018–2022, approx.)

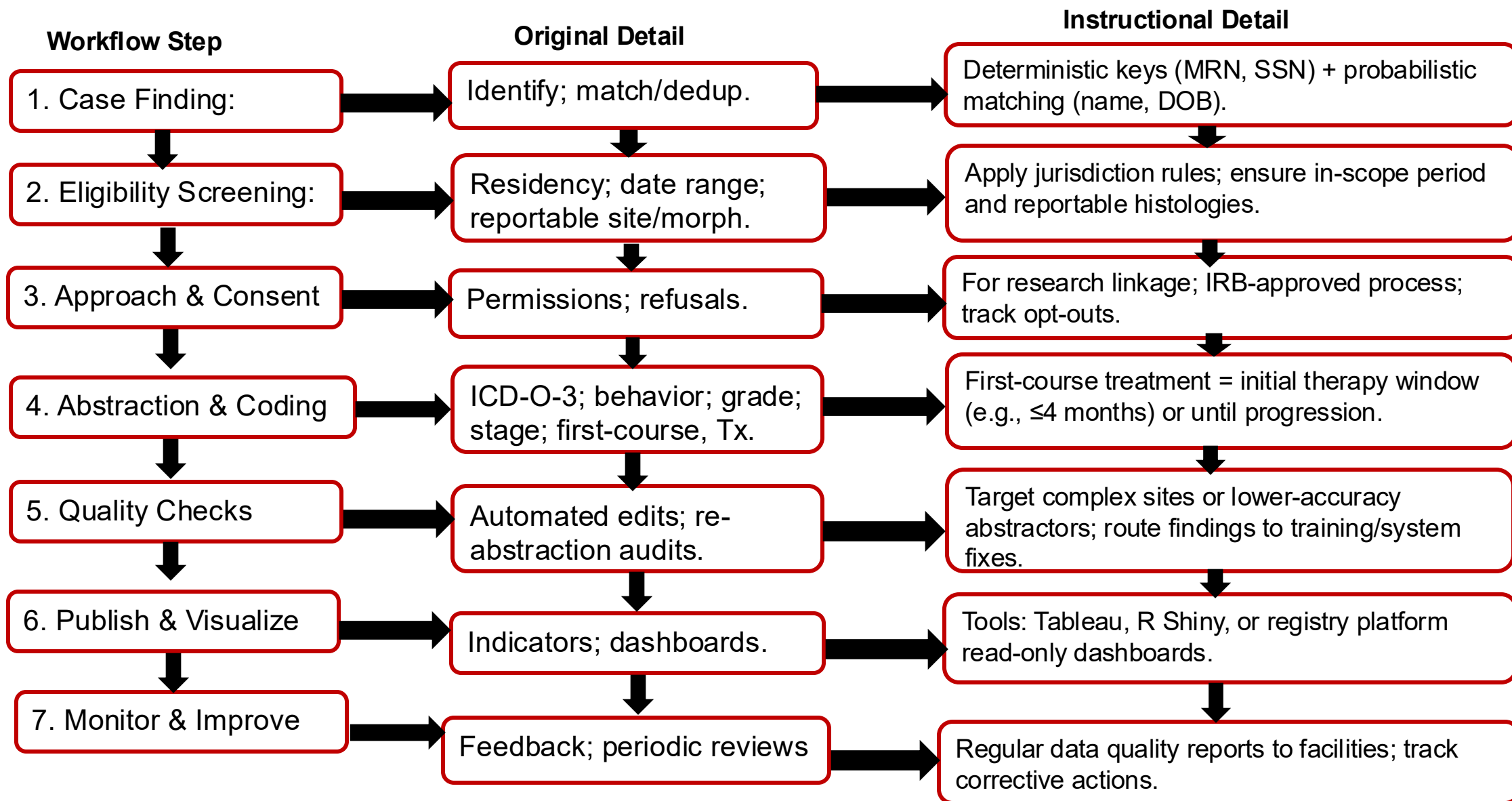


U.S. Female Breast Cancer Stage at Dx (2018–2022): CDC Stat Bite (approx.)

- Core parameters: incidence & mortality (age-adjusted), stage at diagnosis, and survival (overall & relative).
- Evaluation indicators: timeliness (reporting lag), completeness (case capture $\geq 95\%$), accuracy (re-abstraction $\geq 98\%$), consistency (NAACCR edits), and equity (race/ethnicity completeness).

- **Burden metrics quantify the cancer problem**
- **Quality indicators prove the data can be trusted.**

Surveillance Workflow



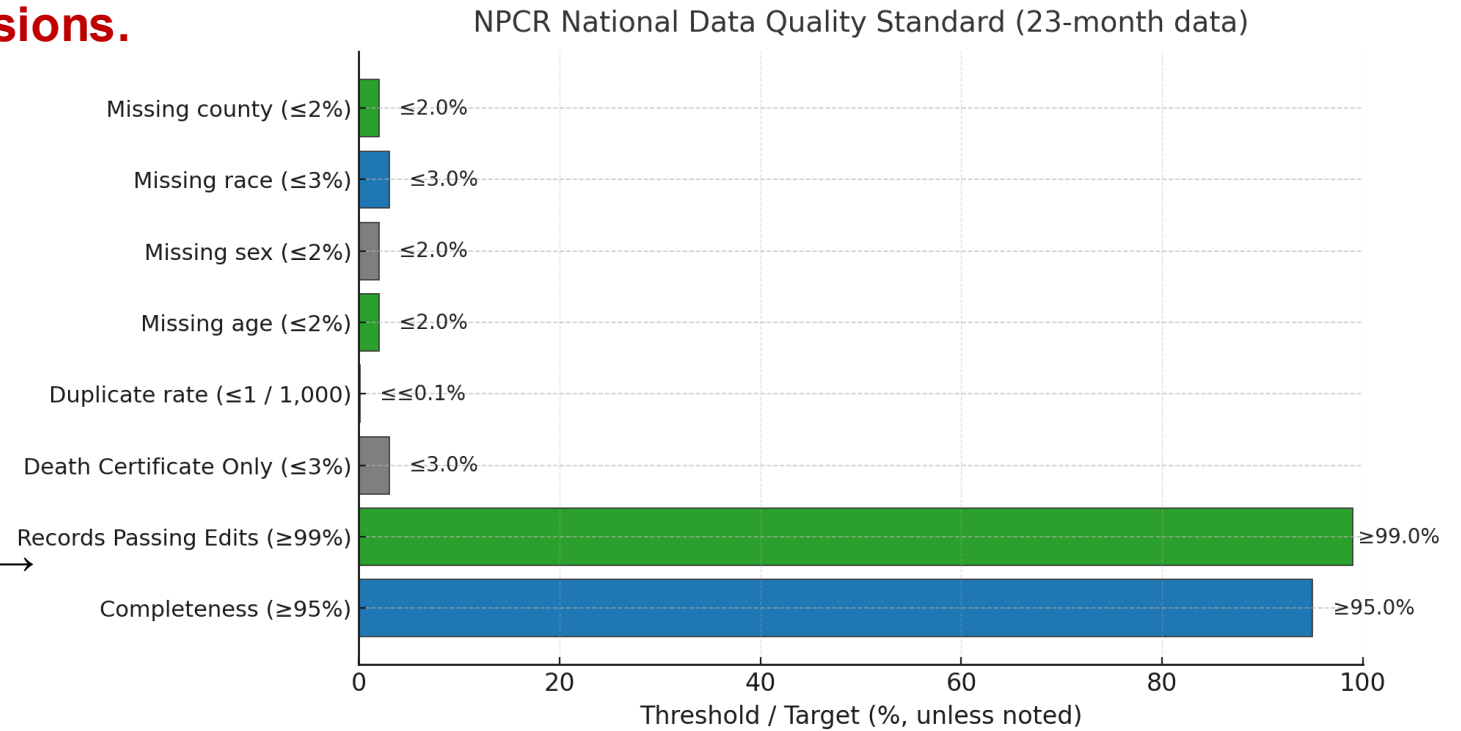
Quality and consent aren't endpoints, they're baked into every step, from case finding to continuous improvement.

Data Quality Framework

Goal: Fitness-for-use that builds trust in incidence, trends, and policy decisions.

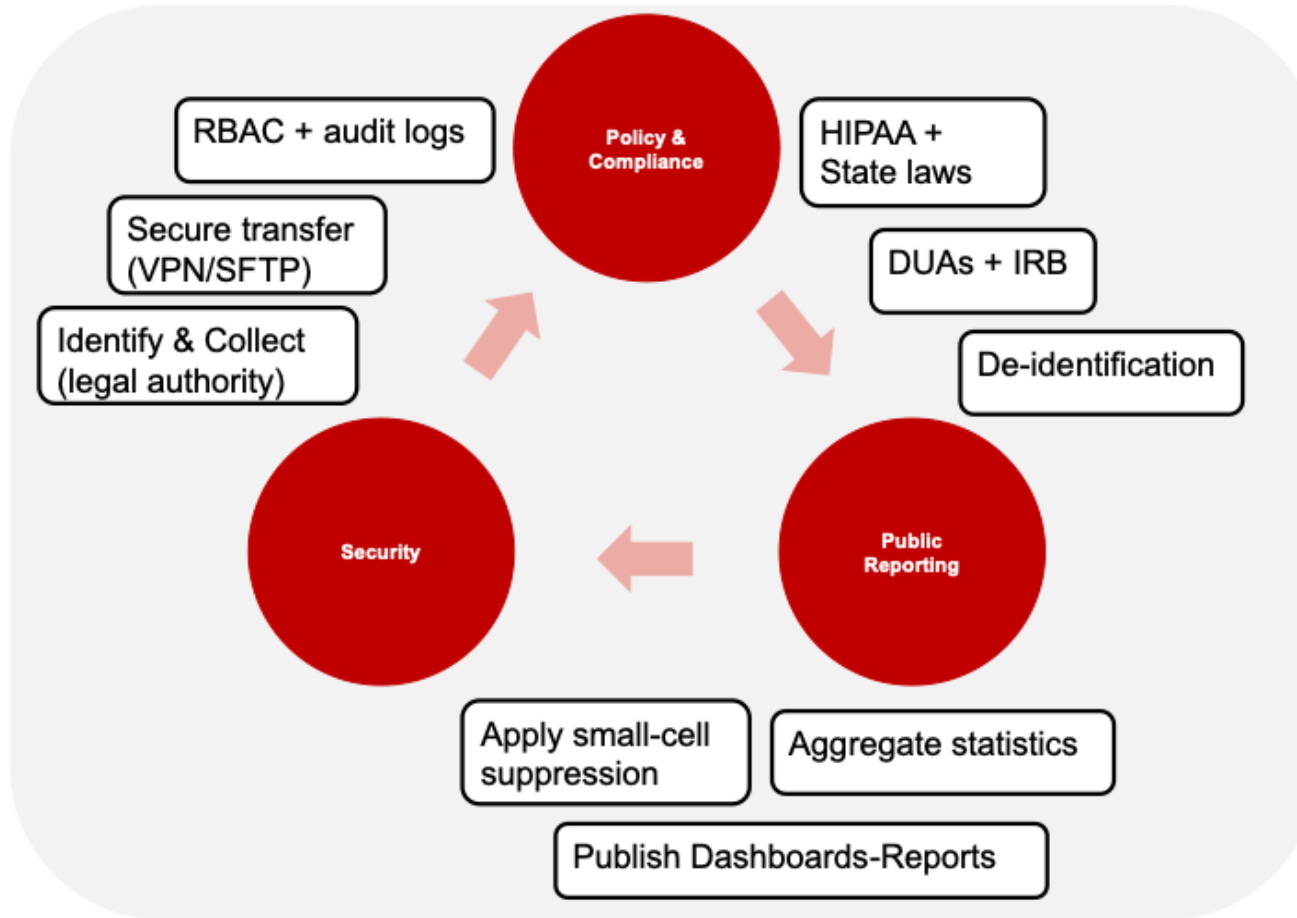
- Dimensions: Completeness, Accuracy, Timeliness, Consistency, Uniqueness
- Checks: Not-null, controlled vocabularies, code edits, cross-record concordance
- Standards: NAACCR edits; SNOMED CT & LOINC for harmonization
- Audits: Re-abstraction, discrepancy review → targeted training/system fixes

No trust, no impact, make data fit-for-use with completeness, accuracy, timeliness, and consistency, every time.



Targets reflect CDC NPCR National Data Quality Standard (Dec 6, 2024): Completeness $\geq 95\%$, Edits $\geq 99\%$, DCO $\leq 3\%$, Duplicate rate $\leq 1/1,000$, Missing Age $\leq 2\%$, Sex $\leq 2\%$, Race $\leq 3\%$, County $\leq 2\%$.

Privacy, Security & Ethical Reporting



Protect Identity, Preserve Insight, Encrypt In, De-Identify Out

Essential Tools & Standards

Coding & Data Structure

- ICD-O-3 (site/morphology)
- AJCC TNM (staging)
- NAACCR Vol II (data dictionary)
- SEER/NAACCR edits

Systems & Interoperability

- Registry platforms (SEER*DMS)
- EHR ↔ Registry
- HL7® FHIR (Condition/Observation)

Analytics & Methods

- Age-standardization
- Joinpoint trend analysis
- Survival (KM, Relative)

- Anchor coding with **ICD-O-3** & **AJCC**; validate with **NAACCR Vol II** and automated edits.
- Use **FHIR** profiles (Condition/Observation; eCR) to move discrete data safely from EHRs to registries.
- Analyze with **Joinpoint** (trend inflection), **age-standardization**, and **survival** metrics to generate comparable, actionable insights.

Standards in, trust out: encode with ICD-O/AJCC, validate with NAACCR, exchange via FHIR, and reveal trends with Joinpoint.

Deliverables & Micro-Checks

What learners submit and how we verify learning in minutes.

Deliverables (turn-in)

- Objective: 1 sentence (population + time)
- Workflow: 6–8 nodes + ≥ 1 QC touchpoint
- Mini-table: Source $\rightarrow$ Field $\rightarrow$ QC $\rightarrow$ Safeguard (≥ 3 rows)
- Privacy list: ≥ 2 safeguards for de-ID

Linked Micro-Checks (in-class)

- MCQ selects correct outcome/measure for a prompt
- Mark QC/edit checks on the workflow diagram
- Pick best source for field X; justify in 1 line
- Identify fields to remove for a given de-ID scenario

Deliverables show the work while micro-checks prove the learning.

Item	Aligns to Outcome	QC Marked	Best Source Chosen	De-ID Safeguards
Objective	✓		✓	
Workflow		✓		
Mini-table		✓	✓	✓
Privacy list				✓

Thank you for your attention!

Please visit my Digital Portfolio Web Page for Modular Details of this Project and other Projects for CPH-705

https://arshah18.github.io/academic-portfolio/coursework/cph-705/cancer-surveillance-modular-analysis/



Rauf Shah
MS, PhD · MPH (Expected 2026)
Lead Health Data Coordinator Level II

Home

About

Public Health Focus

Academic Portfolio

Surveys

Contact

Sitemap

Q Search Rauf Shah's Digital Portfolio

LinkedInGoogle ScholarEmail

Modular Topic/Instructional Analysis — Cancer Surveillance

Modular Topic/Instructional Analysis: Cancer Surveillance

Author: Rauf Shah, MS, Ph.D.
Lead Health Data Coordinator Level II
Division of Oncology and Hematology
University of Nebraska Medical Center, USA
Date: Oct 12, 2025

A structured breakdown for teaching/learning core cancer surveillance concepts and practice.

Modules 1-2: Definition & Purpose

One-liner aims + why it matters; anchors content.

Modules 1-2: Definition & Purpose		
Topic	Original Detail	Instructional Detail
1. Definition	Systematic cancer data tracking. Population-based registries; standard codes. Outputs: burden, stage, survival, and trends.	Emphasize completeness and timeliness. Train on ICD-O-3 (site, morphology) and AJCC (TNM→stage group).
2. Purpose	Define burden & distribution. Track trends & disparities. Support programs, policy, research.	Use concrete disparity examples (late-stage share by race/rurality). Link to policy (targeted screening/navigation).

Teaching Philosophy

My Teaching Journey

In my work with undergraduate and graduate students, I've found that **complex concepts become accessible when grounded in real-world context and practical applications**. Whether guiding students through genomic data analysis or public health surveillance methods, I focus on making technical content relevant and approachable.

Formal Teaching Education

CPH 705 - Public Health Teaching and Training

My graduate training in evidence-based teaching methodologies through CPH 705 provided a systematic framework for developing effective public health education. This course emphasized:

- **Adult Learning Theory:** Applying andragogy principles to public health education
- **Instructional Design:** Systematic approaches from needs assessment to evaluation
- **Multi-Modal Delivery:** Creating accessible learning experiences across different formats
- **Program Evaluation:** Measuring training effectiveness and impact

Teaching Philosophy Papers

Featured: [Combined Teaching Philosophy — Statement \(PDF\)](#)

- Read inline preview

All papers (download links)

- [Draft Teaching Philosophy Statement \(PDF\)](#)
- [Competency/Goal/Standard → Assessment → Learning Objective: Alignment Worksheet \(PDF\)](#)
- [Seven Principles of Adult Learning \(PDF\)](#)
- [Creating a Hook/Intro and Conclusion: Worksheet \(PDF\)](#)
- [Combined Teaching Philosophy: Statement \(PDF\)](#)
- [Teaching Philosophy: Philosophies Worksheet \(PDF\)](#)
- [Teaching Philosophy: Learning Theories Statement \(PDF\)](#)
- [Teaching Philosophy: Learning Theory Worksheet \(PDF\)](#)